Homework 3

Multivariate Statistical Analysis, Fall 2026

Due: November 12, 2026

Submission requirements

- Submit your original handwritten solutions to Questions 1–5 in class. Show the main steps.
- Upload one PDF report for Questions 6–7, together with separate R source files, to the XMU online course system (`course.xmu.edu.cn`). Include your name and student ID on each part, and label answers by question number.
- Use the supplied data and `HW3-data-setup.R`. Cite the data sources and submit the code used to produce your results.
- Submit both parts by the date and submission times announced for this assignment.

Part I: Written Solutions

1. Suppose $\boldsymbol{X} \mid G_k \sim N_2(\boldsymbol{\mu}_k, \Sigma)$, $k = 1, 2$, where

$$\boldsymbol{\mu}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \qquad \boldsymbol{\mu}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

   Use the orientation $\boldsymbol{a} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$.

   (a) Calculate $\boldsymbol{a}$ and the two projected means $\boldsymbol{a}^\top \boldsymbol{\mu}_k$. Briefly explain what the projection is intended to separate.

   (b) With equal prior probabilities and equal misclassification costs, write the linear classification rule and its boundary. Classify $\boldsymbol{x}_A = (2, 1)^\top$ and $\boldsymbol{x}_B = (1, 2)^\top$.

   (c) Keep equal costs but set $(\pi_1, \pi_2) = (0.8, 0.2)$. Find the new boundary, reclassify $\boldsymbol{x}_A$, and explain the direction of its shift relative to part (b).

   (d) Return to equal priors. Let assigning a true G_2 observation to G_1 cost 4, and assigning a true G_1 observation to G_2 cost 1; correct decisions cost zero. Write the cost-sensitive rule and classify $\boldsymbol{x}_B$. Explain whether the discriminant direction or the decision threshold changes. You may use $\log 4 = 1.3863$.

2. Consider the linear model $\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $X \in \mathbb{R}^{n \times k}$ is fixed with full column rank, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, and $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$ with $\sigma^2 > 0$. Let $\hat{\boldsymbol{\beta}} = L\boldsymbol{y}$, where $L = (X^\top X)^{-1} X^\top$. For any linear unbiased estimator $A\boldsymbol{y}$, put $D = A - L$. You may use

$$\mathrm{Cov}(A\boldsymbol{y}) - \mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 D D^\top.$$

   Explain why this establishes the BLUE property in covariance matrix ordering. State whether normal errors are required. You do not need to derive the identity.

3. Consider the multiple linear regression model

$$y_i = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + e_i, \qquad i = 1, \dots, 100,$$

where the predictors are fixed and $e_1, \dots, e_{100}$ are i.i.d. errors with mean zero and unknown variance $\sigma^2 > 0$. Let $\boldsymbol{x}_i = (x_{i1}, x_{i2})^\mathsf{T}$. The observed summaries are

$$\bar{y} = 2, \qquad \bar{\boldsymbol{x}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \sum_{i=1}^{100} \boldsymbol{x}_i \boldsymbol{x}_i^\mathsf{T} = \begin{pmatrix} 200 & 100 \\ 100 & 200 \end{pmatrix},$$

$$\sum_{i=1}^{100} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(y_i - \bar{y}) = \begin{pmatrix} 100 \\ 90 \end{pmatrix}, \qquad \sum_{i=1}^{100} y_i^2 = 560.57.$$

   (a) Write the regression model for the 100 observations in matrix form.

   (b) Find the least squares estimate of $(\beta_0, \beta_1, \beta_2)^\mathsf{T}$.

   (c) Give an unbiased estimate of the error variance σ^2.

   (d) Test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ at the 5% significance level. State the additional distributional assumption needed for an exact t test, and report the test statistic, its null distribution, and your conclusion. You may use $t_{97;\,0.975} = 1.985$, the 0.975 quantile of t_{97}.

4. Consider the multivariate regression model $\boldsymbol{y}_i = B^\mathsf{T}\boldsymbol{x}_i + \boldsymbol{\varepsilon}_i$, where $\boldsymbol{y}_i \in \mathbb{R}^p$, $\boldsymbol{x}_i = (1, x_{i1}, \dots, x_{iq})^\mathsf{T} \in \mathbb{R}^k$, $k = q + 1$, and $B \in \mathbb{R}^{k \times p}$. The predictors are fixed and the errors are independent $N_p(\mathbf{0}, \Sigma)$, with $\Sigma = (\sigma_{j\ell})$ positive definite. Assume $n > k$ and the design matrix X, with rows $\boldsymbol{x}_i^\mathsf{T}$, has full column rank.

   (a) Write the model for all n observations in matrix form, giving the dimensions, and write down the least squares estimator $\hat{B}$.

   (b) Let $\hat{R} = Y - X\hat{B}$ and $E = \hat{R}^\mathsf{T}\hat{R}$. You may use the identity

$$(Y - XC)^\mathsf{T}(Y - XC) = E + (C - \hat{B})^\mathsf{T} X^\mathsf{T} X (C - \hat{B}), \qquad C \in \mathbb{R}^{k \times p}.$$

   Briefly explain why the extra term is positive semidefinite and what this implies about residual sums of squares for fixed response combinations $Y\boldsymbol{a}$, $\boldsymbol{a} \in \mathbb{R}^p$. You do not need to derive the identity.

   (c) Derive the distribution of each column of $\hat{B}$, and the covariance structure between its columns.

5. Suppose $\boldsymbol{x}_i \in \mathbb{R}^p$ belong to two classes of sizes $n_1, n_2 \geq 2$, with $n = n_1 + n_2$. Code $y_i = -n/n_1$ in class 1 and $y_i = n/n_2$ in class 2. Let $\boldsymbol{\delta} = \bar{\boldsymbol{x}}_2 - \bar{\boldsymbol{x}}_1 \neq \mathbf{0}$, and define

$$W = \sum_{g=1}^{2} \sum_{i \in G_g} (\boldsymbol{x}_i - \bar{\boldsymbol{x}}_g)(\boldsymbol{x}_i - \bar{\boldsymbol{x}}_g)^\mathsf{T}, \qquad S_{\text{pl}} = \frac{W}{n-2}.$$

Assume S_{pl} is nonsingular. Show that the least squares slope in $y_i = \beta_0 + \boldsymbol{x}_i^\mathsf{T}\boldsymbol{\beta} + \varepsilon_i$ is proportional to the LDA direction $S_{\text{pl}}^{-1}\boldsymbol{\delta}$.

Hints: With $\bar{\boldsymbol{x}}$ the overall feature mean, $w = n_1 n_2 / n$, and $T = \sum_i (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^\mathsf{T}$, you may use

$$\bar{y} = 0, \qquad \sum_i (\boldsymbol{x}_i - \bar{\boldsymbol{x}}) y_i = n\boldsymbol{\delta}, \qquad T = W + w\boldsymbol{\delta}\boldsymbol{\delta}^\mathsf{T}.$$

Center the slope normal equations and examine $TS_{\text{pl}}^{-1}\boldsymbol{\delta}$. An explicit proportionality constant or a matrix-inverse identity is not required.

Part II: Data Analysis and Code

Run `source("HW3-data-setup.R")` from the supplied folder. Use the fixed training/test splits. Fit models using training data only and evaluate on the test set. Present concise tables, figures and interpretation; submit R code separately.

6. **LDA for banknote authentication (R).** Use the UCI Banknote Authentication data supplied in `data/data_banknote_authentication.txt`. The four predictors are `variance`, `skewness`, `kurtosis`, and `entropy`; the response is `class`, with levels 0 and 1. The setup script removes exact duplicate rows and creates `bn_train` (943 rows) and `bn_test` (405 rows).

   (a) Report the class counts in both splits and give a small table of training class means for the four predictors.

   (b) Fit `MASS::lda` with all four predictors and equal priors $(0.5, 0.5)$ and equal misclassification costs. Report the discriminant coefficients and draw one labelled plot of the training discriminant scores by class. Briefly describe the separation and explain why there is only one discriminant axis.

   (c) Predict classes for the test set. Give a confusion matrix (clearly label true and predicted classes) and the test error rate. In a few sentences, interpret the result and state one limitation of the analysis.

   (d) Refit LDA with priors $(0.8, 0.2)$ in class order $(0, 1)$, keeping all four predictors and equal costs. On the same test set, report how many predicted labels change from part (c). Explain which class is favoured and why.

7. **Regression for concrete mixtures (R).** Use the UCI Concrete Slump Test data in `data/slump_test.data`. Each row is a concrete mixture. The setup script creates `concrete_train` (72 rows) and `concrete_test` (31 rows). Strength is measured in MPa at 28 days; ingredient quantities are in kg per m^3. For parts (a)–(c), use the following prescribed model:

   `strength` $\sim$ `cement + water + fly_ash`.

   (a) Fit the model on the training set. Report a small table of coefficient estimates and standard errors, and R^2. Interpret the water coefficient conditional on the other two ingredients. Give its 95% confidence interval and the conclusion of a two-sided test at 5%.

   (b) Using the training fit, draw residual-versus-fitted and normal Q–Q plots (one two-panel figure is sufficient). Use standardized residuals in the Q–Q plot. State one finding and one assumption or limitation relevant to the regression conclusions. Formal diagnostic tests are not required.

   (c) Predict strength for the test set and report test RMSE. Compare it with a baseline that predicts the training mean strength for every test row. Briefly explain what the comparison suggests. Here RMSE $= \{n_{\text{test}}^{-1} \sum_{i \in \text{test}} (y_i - \hat{y}_i)^2\}^{1/2}$.

   (d) On the same training set, fit `cbind(slump, flow) ~ cement + water + fly_ash`. Slump and flow are measured in cm. Report the residual covariance estimate $S_e = R^{\mathsf{T}} R/(n-k)$ and the residual correlation between slump and flow, where R is the residual matrix and k counts coefficients per response, including the intercept. Briefly interpret the residual correlation. No joint test or prediction interval is required.